

Stimuli-Responsive Anion Transport through Acylhydrazone-Based Synthetic Anionophores

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Cite This: *Org. Lett.* 2021, 23, 7319–7324



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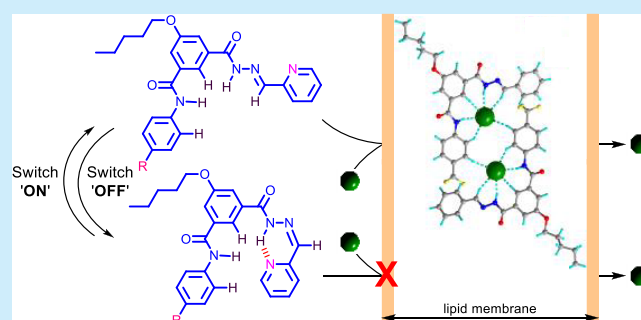


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Supporting Information

ABSTRACT: Photoswitchable acylhydrazone-based synthetic anionophores are reported. Single-crystal X-ray structure and ^1H NMR titration studies confirmed the chloride binding in solid and solution states. The ion transport activity of **1a** was greatly attenuated through a phototriggered *E* to *Z* photoisomerization process, and the photoisomerized deactivated state showed high kinetic stability due to an intramolecular hydrogen bond. Switchable “OFF–ON” transport activity was achieved by the application of light and acid-catalyzed reactivation process.



The ion transport across the cell membrane mediated through natural proteins (channels, carriers, and pumps)^{1–3} is important for various biological functions.^{4,5} Many of these natural ion transporters are “gated”, a behavior through which they are switched “ON” and “OFF”, induced through the stimuli of light, pH, voltage, or ligand, etc.^{6–8} A wide variety of artificial mimics, in the form of ion channels and mobile carriers, has been reported.^{9–16} Supramolecular mobile carriers and ion channels have attracted significant interest because of their diverse chemical biology¹⁷ and therapeutic applications.^{12,15,18–24}

A wide variety of synthetic “gated” ion transport systems has been generated that involve pH, voltage, light, ligand, etc., as external stimuli.^{25–32} In particular, light-responsive ionophores are attractive targets due to the tunability of light intensity and high spatiotemporal control.³³ Light-responsive ionophores mostly involve either photoswitchable (e.g., azobenzene,^{34–39} spiropyran,⁴⁰ etc.) or photocleavable (e.g., *o*-nitro aromatics)^{41,42} groups. Azobenzene, because of its short-responsive nature, is one of the most commonly used photoswitches.⁴³ However, spontaneous *cis* to *trans* thermal relaxation is one of the main drawbacks of this photoswitch. In this regard, acylhydrazone-based photoswitches are highly tunable and can be controlled by applying light, redox, thermal, or pH changes.^{44–46} Especially, (*E*)-acylhydrazones, which are based on 2-pyridinecarboxaldehyde, upon photoisomerization of the C=N double bond give the (*Z*)-form having more thermal stability due to intramolecular hydrogen bond formation.^{47–49} Therefore, acylhydrazones have been well explored in different fields, e.g., ion-pair recognition,⁵⁰ supramolecular self-assemblies,⁵¹ photoregulated molecular shuttles,⁵² etc. In 2017, Zhou et al. introduced acylhydrazones

as a switch to design a crown ether-based cation-selective ion channel.⁵³ The self-assembled channel structure was disassembled upon photoirradiation at 300 nm and then regenerated by 365 nm of photoirradiation, leading to reversible “OFF–ON” channel activity. We realized that 2-pyridinecarboxaldehyde based acylhydrazone system could be easily tuned to design a stimulus-responsive anion transporter by creating a cavity that can offer multivalent and strong anion recognition. Although there is a cation binding domain in the receptor, the site may not bind and transport cations, as this ion would be exposed to the hydrophobic domains of the lipid membrane. On the basis of this understanding, we designed a class of acylhydrazone-based anion carriers **1a–1d** (Figure 1). Our motif contains 5-(pentyloxy)isophthalic biscarboxamide as a cavity for strong anion recognition and 2-pyridyl acylhydrazone as a stimuli-responsive switch. Photoisomerization from (*E*)-form to (*Z*)-form across the C=N double bond using electromagnetic radiation would block the anion binding site due to intramolecular hydrogen bonding. On the other hand, the inactive (*Z*)-form can be converted back to the (*E*)-form through acid-catalyzed reactivation process. Molecule **1d**, which has a phenyl ring instead of a pyridyl ring, was not expected to be an efficient stimuli-responsive switch. Different aromatic moieties were connected to the amide linkage to fine-

Received: July 6, 2021

Published: September 14, 2021



ACS Publications

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7319

<https://doi.org/10.1021/acs.orglett.1c02249>
Org. Lett. 2021, 23, 7319–7324

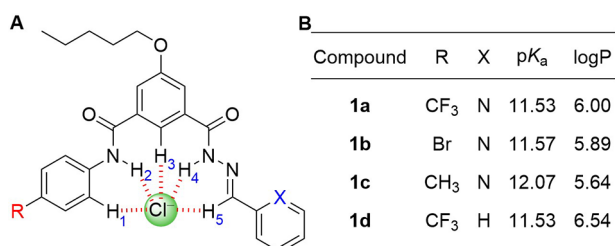
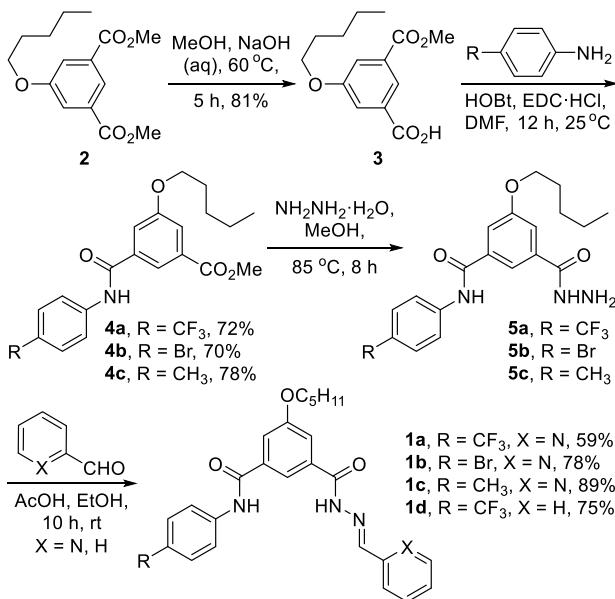


Figure 1. Chemical structures (A), pK_a (N–H₂) and logP values (B) of compounds 1a–1d.

tune the binding affinity and lipophilicity of the molecules.⁵⁴ The pK_a and logP values were calculated using Marvin Sketch Program (Figure 1B).⁵⁵

The synthesis of the desired compounds 1a–1d was achieved from compound 2, which was prepared from 5-hydroxyisophthalic acid following a reported protocol.^{56,57} Monohydrolysis of compound 2 using sodium hydroxide furnish compound 3, which upon coupling with different aromatic amines using EDC·HCl gave monoamide derivatives 4a–4c in excellent yield (Scheme 1). These monoamide ester

Scheme 1. Synthesis of Desired Anionophores 1a–1d



derivatives, upon reacting with hydrazine hydrate followed by coupling with pyridine-2-carboxaldehyde, furnished the desired final compounds 1a–1c in excellent yields. On the other hand, compound 1d was synthesized by reacting 4a with hydrazine hydrate followed by coupling with benzaldehyde.

Initial evidence of the anion binding of these anionic receptors was obtained from ¹H NMR studies in acetonitrile-*d*₃. Upon titration with TBACl in acetonitrile-*d*₃, significant shifts of N–H₁, N–H₂, C_{Ar}–H₃, C_{Ar}–H₄, and C_{imine}–H₅ were observed, which indicated the involvement of these protons in an overall anion binding process through N–H···Cl[−] and C–H···Cl[−] hydrogen bonding interactions (Figures S1, S7, S9, S11). ¹H NMR titration data of 1a with TBACl is shown the Figure 2. Further binding analysis using the BindFit program^{58,59} (Figures S2, S8, S10, S12) revealed 1:1 binding stoichiometry with the binding affinity sequence of 1a > 1b > 1d > 1c, and the binding constants (K_{a(1:1)}/Cl[−]) obtained are

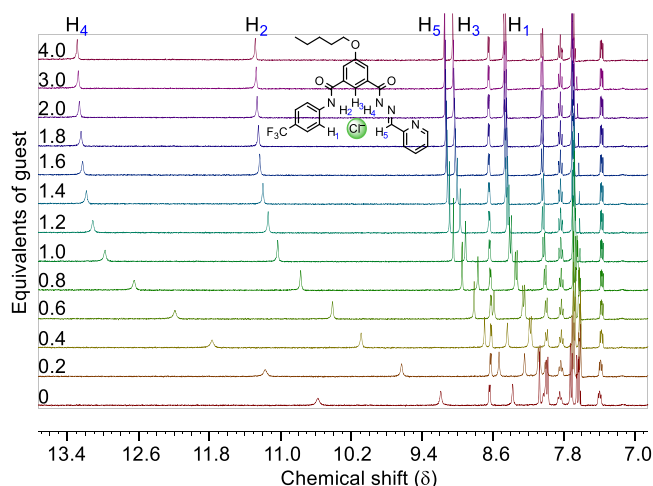


Figure 2. ¹H NMR titration of 1a with TBACl.

11168 ± 675 M^{−1}, for 1a, 7934 ± 573 M^{−1} for 1b, 4201 ± 257 M^{−1} for 1c, and 6037 ± 385 M^{−1} for 1d, respectively (Table S1). This sequence is an outcome of the pK_a values of these receptors. Binding studies of 1a were also done with Br[−], I[−], and ClO₄[−] anions respectively (Figures S3, S5, and S13), and the study provided K_{a(1:1)}/Br[−] = 2596 ± 289 M^{−1} and K_{a(1:1)}/I[−] = 69 ± 17 M^{−1} (Figures S4 and S6). No change in the chemical shift was observed upon the addition of ClO₄[−] anion. The observed binding sequence, Cl[−] > Br[−] > I[−] > ClO₄[−], may be attributed to the high charge density of Cl[−] and its perfect fitting to the binding cavity of the receptor. Furthermore, 1a in its (Z)-form showed a weak Cl[−] binding with TBACl (Figure S14), with the association constant of K_{a(1:1)}/Cl[−] = 53 ± 11 M^{−1} (Figure S15), which indicates the blockage of the binding site due to photoisomerization across C=N bond. The direct evidence of chloride binding of 1a was obtained from the electrospray ionization mass spectrometric (ESI-MS) study. The mass spectrometric data provided peaks at *m/z* = 533.1575 and 535.1554, which correspond to the [1a + Cl[−]] complex in the solution state (Figures S16).

Among all synthesized compounds, cocrystals were obtained for only 1d with TBACl in acetonitrile. The single-crystal X-ray diffraction analysis revealed a (1d)₂(Cl[−])₂ binding mode. In the cavity created by two molecules of 1d, two Cl[−] ions were encapsulated, and each anion formed five hydrogen bonds with one molecule and one hydrogen bond with the second one (Figures 3A, S40, S4, S42, and S43). The distance between two Cl[−] ions was 5.672 Å, which indicates no repulsion between these anions.

The ion transport activity of 1a–1d was evaluated across large unilamellar vesicles (LUVs) entrapped with 8-hydroxy-1,3,6-pyrenetrisulfonate (HPTS) using egg-yolk phosphatidylcholine (EYPC) lipid buffered at pH_{in} = 7.0 using 10 mM of HEPES and 100 mM NaCl (please see Supporting Information).⁶⁰ A pH gradient, ΔpH = 0.8 (pH_{in} = 7.0 and pH_{out} = 7.8), was applied by adding sodium hydroxide in the extravesicular buffer. The equilibration of the applied pH gradient by either influx of OH[−] ions or efflux of H⁺ ions, in the presence of each compound, was monitored by measuring the fluorescence intensity of HPTS at λ = 510 nm (λ_{ex} = 450 nm). Finally, vesicles were lysed using triton-100 to get 100% activity. All of the ionophores 1a–1d showed a significant increment in the fluorescent intensity with the activity sequence of 1d > 1a > 1b > 1c (Figure 3B). This sequence

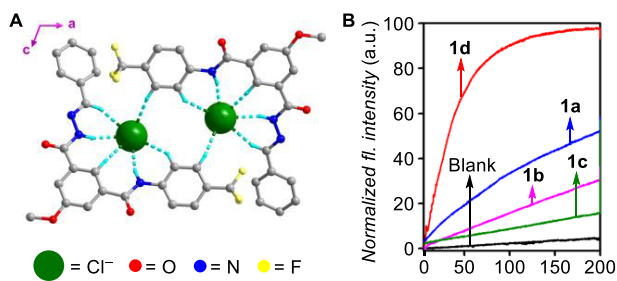


Figure 3. X-ray crystal structure of compound **1d** complexed with Cl^- ion in a 2:2 (**1d**: Cl^-) binding pattern (A). Note: The tetrabutylammonium cations, hydrogen atoms, C_4H_8 units were omitted for clarity. Activity comparison of **1a**–**1d** (0.3 μM each) across EYPC–LUVs > HPTS (B).

is explained on the basis of the collective involvement of binding affinity and membrane permeability of the anionophores. Hill analysis furnish the EC_{50} values of 0.62, 1.58, 2.08, and 0.11 μM for **1a**, **1b**, **1c**, and **1d**, respectively (Figure S18–S21). The Hill coefficient of $n \sim 2$ is indicative of the involvement of two molecules in generating the active ion transport system, and data is in synergy to that observed in the crystal structure of **1d** with Cl^- ion. Furthermore, changing the extravesicular anions using different NaX ($\text{X}^- = \text{Cl}^-$, Br^- , I^- , ClO_4^- , and NO_3^-) salts, the activity sequence of $\text{Cl}^- > \text{Br}^- > \text{NO}_3^- > \text{I}^- > \text{ClO}_4^-$ was obtained (Figure 4B), suggesting the involvement of anions in the transport process. The above sequence may be attributed to the combinative effects of the

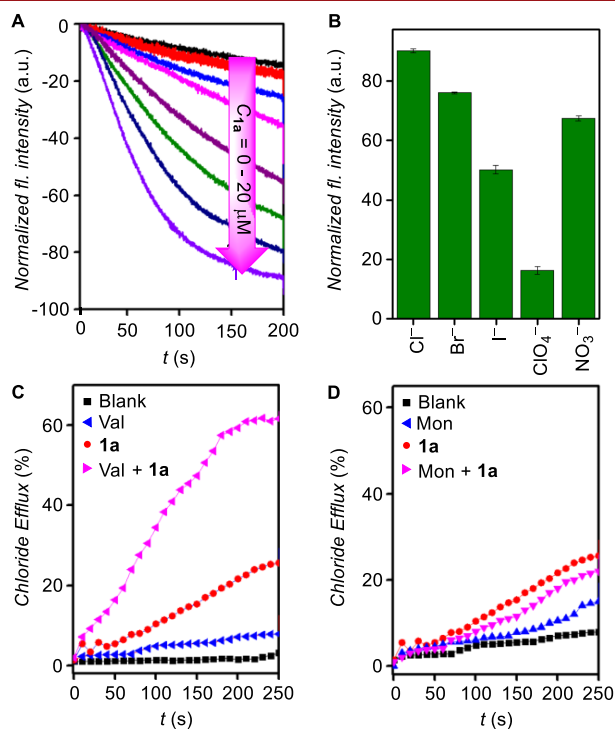


Figure 4. Concentration-dependent activity of compound **1a** across EYPC–LUVs > Lucigenin (A). Anion selectivity of **1a** (0.3 μM) by varying external anions across EYPC–LUVs > HPTS; each bar graph represents mean ion transport activity, calculated from three independent experiments (B). Normalized chloride efflux of **1a** in the presence and absence of valinomycin (C), and the presence and absence of monensin (D).

anion binding affinity and membrane permeability of the transporting molecules.

Subsequently, the influx of Cl^- ions by **1a**–**1d** was monitored across LUVs, entrapping a lucigenin dye and 200 mM NaNO_3 (see SI).⁶¹ Subsequently, NaCl pulse was applied in the extravesicular solution to create a $\text{Cl}^-/\text{NO}_3^-$ gradient across the vesicles. The influx of Cl^- ions was monitored by measuring the fluorescence intensity at $\lambda = 455$ nm ($\lambda_{\text{ex}} = 535$ nm, for lucigenin). The dose–response Cl^- influx activity of **1a** and **1d** are shown in Figures 4A and S24. Hill analysis furnished EC_{50} values of 5.70 and 0.50 μM for **1a** and **1d**, respectively (Figure S23 and S24). The dose-dependent Cl^- influx activities of **1b** and **1c** are shown in Figure S25. Hill analysis could not be done with these compounds due to the precipitation issues at higher concentrations. Furthermore, no change in the transport activity was observed by changing the extravesicular cations ($\text{M}^+ = \text{Li}^+$, Na^+ , K^+ , Rb^+ , and Cs^+), which rules out any involvement of cations in the transport process (Figure S27).

To get further mechanistic insights for ion transport, chloride efflux from LUVs by **1a** in the presence and absence of either valinomycin (a highly selective K^+ transporter) or monensin (a H^+/K^+ antiporter) was monitored by a chloride ion selective electrode (ISE). With intravesicular KCl (300 mM) and extravesicular potassium gluconate, a significant enhancement in the Cl^- efflux rate was observed for **1a** in the presence of valinomycin compared to that with **1a** alone (Figure 4C), and on the other hand, no significant change in the Cl^- efflux was observed in the presence of monensin. The KCl efflux only under electrogenic potassium transporter is the evidence of an electrogenic chloride transport by **1a**.^{23,62,63} Furthermore, the U-tube experiment confirmed the carrier mode of ion transport (Figure S30).

Photoisomerization studies of the compounds **1a**–**1d** were monitored by UV–vis absorption and ^1H NMR spectroscopy. UV band at 300 nm corresponding to π – π^* transition got substantially decreased upon irradiation with 312 nm light, and the N–H proton of (*E*)-acylhydrazone shifted significantly to the downfield region upon photoisomerization to the (*Z*)-form (Figure S31–S34).⁴⁸ For **1a**–**1c**, the photoconversion was quantitative and **1a** was seen to be stable toward thermal isomerization for several days (Figure S35). This excellent thermal stability is a consequence of an intramolecular hydrogen bonding between hydrazine N–H and pyridine N in the photoisomerized *Z* state. The (*Z*)-form of **1d** that lacks such intramolecular hydrogen bonding was photoisomerized only up to 38%, and complete relaxation of the (*Z*)-form was observed within a day while keeping in the dark (Figure S36). Reversible isomerization was achieved by the application of light (312 nm) to convert *E* to *Z* form, followed by the addition of triflic acid (TFA, 5 equiv) to regain the *E* form. Triethylamine (TEA, 5 equiv) was added to neutralize the protonated form after 30 min of acid treatment (Figure S37). This process of light induced photoisomerization and acid catalyzed reactivation process was monitored over a number of repeated cycles without any loss of efficiency (Figure 5A).

Photoirradiation of **1a** in acetonitrile at 312 nm significantly reduced its transport activity due to the steric hindrance in the binding site, generated by *E* to *Z* photoisomerization (Figure S38A). However, no significant change in the transport activity was observed while keeping the photoisomerized sample of **1a** in the dark for 9 h due to its high thermal stability in this photoisomerized *Z* state (Figure S38B). Reversible switching

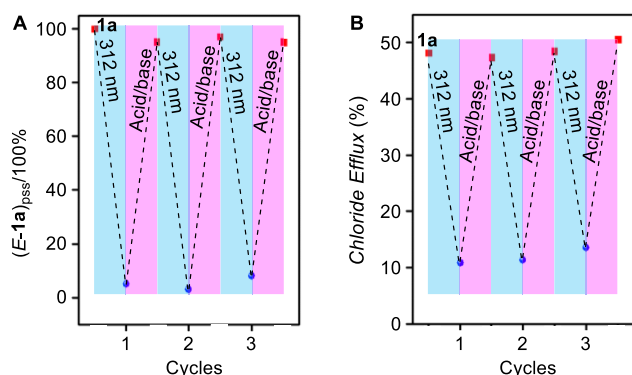


Figure 5. Reversible *E* to *Z* switching of **1a** upon irradiation at 312 nm and acid/base treatment (5 equiv each) (A). Chloride efflux of **1a** in *E* form, after irradiation at 312 nm and after acid/base treatment (5 equiv) (B).

of ion transport activity using different stimuli of light and acid/base treatment were studied by monitoring the chloride efflux from LUVs. There was 82% of decrease in the chloride efflux upon photoirradiation of **1a** as acetonitrile solution, which was efficiently regained back by applying acid/base treatment to the photoisomerized solution of **1a** (Figure S39). This process of deactivation and reactivation was successfully carried out for three repeated cycles without the loss of efficiency (Figure 5B).

In conclusion, we have demonstrated photoresponsive anionophoric activities of acylhydrazones **1a–1c**. Chloride binding by the *E*-form of the system was confirmed by single crystal X-ray diffraction and ¹H NMR spectroscopic studies. The most active transporter **1a** provided EC₅₀ = 0.62 μM and Hill coefficient ~2, indicating the involvement of two molecules in the formation of active transporter. The mechanistic studies confirmed a carrier-mediated anion antiport as the primary transport process. The gated ion transport by **1a** was established through the reversible *E–Z* isomerization.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.1c02249>.

General methods, experimental methods, NMR spectra, and crystal structure (PDF)

Accession Codes

CCDC 2092373 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was funded by SERB, Govt. of India (EMR/2016/001897), and DST, Govt. of India (Under Indo-Russian Collaborative Project Scheme, DST/INT/RUS/RSF/P-24). MA and DM thank UGC, Govt. of India, for research fellowships. SC thanks Prime Minister's Research Fellowship.

■ REFERENCES

- (1) Lichen, S.; Riccardis, F.; Izzo, I.; Tecilla, P. Artificial Anion Transporters in Bilayer Membranes. *Curr. Drug Discovery Technol.* **2008**, *5*, 86–97.
- (2) Jentsch, T. J.; Hübner, C. A.; Fuhrmann, J. C. Ion channels: function unravelled by dysfunction. *Nat. Cell Biol.* **2004**, *6*, 1039–1047.
- (3) Gale, P. A.; Light, M. E.; McNally, B.; Navakhun, K.; Sliwinski, K. E.; Smith, B. D. Co-transport of H⁺/Cl[−] by a synthetic prodigiosin mimic. *Chem. Commun.* **2005**, 3773–3775.
- (4) Chiu, S. Y.; Wilson, G. F. The role of potassium channels in Schwann cell proliferation in Wallerian degeneration of explant rabbit sciatic nerves. *J. Physiol.* **1989**, *408*, 199–222.
- (5) Nowak, L.; Bregestovski, P.; Ascher, P.; Herbet, A.; Prochiantz, A. Magnesium gates glutamate-activated channels in mouse central neurones. *Nature* **1984**, *307*, 462–465.
- (6) Nagel, G.; Ollig, D.; Fuhrmann, M.; Kateriya, S.; Musti, A. M.; Bamberg, E.; Hegemann, P. Channelrhodopsin-1: A Light-Gated Proton Channel in Green Algae. *Science* **2002**, *296*, 2395–2398.
- (7) Catterall, W. A. Structure and functions of Voltage-gated ion channels. *Annu. Rev. Biochem.* **1995**, *64*, 493–531.
- (8) Jaiteh, M.; Taly, A.; Hénin, J. Evolution of Pentameric Ligand-Gated Ion Channels: Pro-Loop Receptors. *PLoS One* **2016**, *11*, e0151934.
- (9) Ko, S.-K.; Kim, S. K.; Share, A.; Lynch, V. M.; Park, J.; Namkung, W.; Van Rossom, W.; Busschaert, N.; Gale, P. A.; Sessler, J. L.; Shin, I. Synthetic ion transporters can induce apoptosis by facilitating chloride anion transport into cells. *Nat. Chem.* **2014**, *6*, 885–892.
- (10) Winstanley, K. J.; Allen, S. J.; Smith, D. K. Encapsulated binding sites—synthetically simple receptors for the binding and transport of HCl. *Chem. Commun.* **2009**, 4299–4301.
- (11) Talukdar, P.; Bollot, G.; Mareda, J.; Sakai, N.; Matile, S. Synthetic Ion Channels with Rigid-Rod π -Stack Architecture that Open in Response to Charge-Transfer Complex Formation. *J. Am. Chem. Soc.* **2005**, *127*, 6528–6529.

- (12) Saha, T.; Gautam, A.; Mukherjee, A.; Lahiri, M.; Talukdar, P. Chloride Transport through Supramolecular Barrel-Rosette Ion Channels: Lipophilic Control and Apoptosis-Inducing Activity. *J. Am. Chem. Soc.* **2016**, *138*, 16443–16451.
- (13) Mondal, D.; Sathyan, A.; Shinde, S. V.; Mishra, K. K.; Talukdar, P. Tripodal cyanurates as selective transmembrane Cl^- transporters. *Org. Biomol. Chem.* **2018**, *16*, 8690–8694.
- (14) Shinde, S. V.; Talukdar, P. Transmembrane H^+/Cl^- cotransport activity of bis(amido)imidazole receptors. *Org. Biomol. Chem.* **2019**, *17*, 4483–4490.
- (15) Saha, T.; Hossain, M. S.; Saha, D.; Lahiri, M.; Talukdar, P. Chloride-Mediated Apoptosis-Inducing Activity of Bis(sulfonamide) Anionophores. *J. Am. Chem. Soc.* **2016**, *138*, 7558–7567.
- (16) Valkenier, H.; Judd, L. W.; Li, H.; Hussain, S.; Sheppard, D. N.; Davis, A. P. Preorganized Bis-Thioureas as Powerful Anion Carriers: Chloride Transport by Single Molecules in Large Unilamellar Vesicles. *J. Am. Chem. Soc.* **2014**, *136*, 12507–12512.
- (17) Steller, L.; Kreir, M.; Salzer, R. Natural and artificial ion channels for biosensing platforms. *Anal. Bioanal. Chem.* **2012**, *402*, 209–230.
- (18) Shen, B.; Li, X.; Wang, F.; Yao, X.; Yang, D. A synthetic chloride channel restores chloride conductance in human cystic fibrosis epithelial cells. *PLoS One* **2012**, *7*, e34694.
- (19) Gokel, G. W.; Carasel, I. A. Biologically active, synthetic ion transporters. *Chem. Soc. Rev.* **2007**, *36*, 378–389.
- (20) Akhtar, N.; Biswas, O.; Manna, D. Biological applications of synthetic anion transporters. *Chem. Commun.* **2020**, *56*, 14137–14153.
- (21) Carreira-Barral, I.; Mielczarek, M.; Alonso-Carrillo, D.; Capurro, V.; Soto-Cerrato, V.; Pérez Tomás, R.; Caci, E.; García-Valverde, M.; Quesada, R. Click-tambjamines as efficient and tunable bioactive anion transporters. *Chem. Commun.* **2020**, *56*, 3218–3221.
- (22) Van Rossom, W.; Asby, D. J.; Tavassoli, A.; Gale, P. A. Perenosins: a new class of anion transporter with anti-cancer activity. *Org. Biomol. Chem.* **2016**, *14*, 2645–2650.
- (23) Chen, L.; Berry, S. N.; Wu, X.; Howe, E. N. W.; Gale, P. A. Advances in Anion Receptor Chemistry. *Chem.* **2020**, *6*, 61–141.
- (24) Li, H.; Valkenier, H.; Judd, L. W.; Brotherhood, P. R.; Hussain, S.; Cooper, J. A.; Jurček, O.; Sparkes, H. A.; Sheppard, D. N.; Davis, A. P. Efficient, non-toxic anion transport by synthetic carriers in cells and epithelia. *Nat. Chem.* **2016**, *8*, 24–32.
- (25) Kobuke, Y.; Morita, K. Artificial ion channels of amphiphilic ion pairs composed of oligoether-ammonium and hydrophobic carboxylate or phosphates. *Inorg. Chim. Acta* **1998**, *283*, 167–174.
- (26) Fyles, T. M.; Loock, D.; Zhou, X. A Voltage-Gated Ion Channel Based on a Bis-Macrocyclic Bolaamphiphile. *J. Am. Chem. Soc.* **1998**, *120*, 2997–3003.
- (27) Sakai, N.; Matile, S. Synthetic multifunctional pores: lessons from rigid-rod β -barrels. *Chem. Commun.* **2003**, 2514–2523.
- (28) Tedesco, M. M.; Ghebremariam, B.; Sakai, N.; Matile, S. Modeling the Selectivity of Potassium Channels with Synthetic, Ligand-Assembled π Slides. *Angew. Chem., Int. Ed.* **1999**, *38*, 540–543.
- (29) O'Shea, E.; Rutkowski, R.; Stafford, W.; Kim, P. Preferential heterodimer formation by isolated leucine zippers from fos and jun. *Science* **1989**, *245*, 646–648.
- (30) Wilson, C. P.; Webb, S. J. Palladium(ii)-gated ion channels. *Chem. Commun.* **2008**, 4007–4009.
- (31) Zhou, Y.; Chen, Y.; Zhu, P.-P.; Si, W.; Hou, J.-L.; Liu, Y. Reversible photo-gated transmembrane channel assembled from an acylhydrazone-containing crown ether triad. *Chem. Commun.* **2017**, *53*, 3681–3684.
- (32) Reyes, R.; Duprat, F.; Lesage, F.; Fink, M.; Salinas, M.; Farman, N.; Lazdunski, M. Cloning and expression of a novel pH-sensitive two pore domain K^+ channel from human kidney. *J. Biol. Chem.* **1998**, *273*, 30863–30869.
- (33) Banghart, M. R.; Volgraf, M.; Trauner, D. Engineering Light-Gated Ion Channels. *Biochemistry* **2006**, *45*, 15129–15141.
- (34) Jog, P. V.; Gin, M. S. A Light-Gated Synthetic Ion Channel. *Org. Lett.* **2008**, *10*, 3693–3696.
- (35) Yang, R.-Y.; Bao, C.; Lin, Q.-N.; Zhu, L.-Y. A light-regulated synthetic ion channel constructed by an azobenzene modified hydrophile. *Chin. Chem. Lett.* **2015**, *26*, 851–856.
- (36) Liu, T.; Bao, C.; Wang, H.; Lin, Y.; Jia, H.; Zhu, L. Light-controlled ion channels formed by amphiphilic small molecules regulate ion conduction via *cis-trans* photoisomerization. *Chem. Commun.* **2013**, *49*, 10311–10313.
- (37) Choi, Y. R.; Kim, G. C.; Jeon, H.-G.; Park, J.; Namkung, W.; Jeong, K.-S. Azobenzene-based chloride transporters with light-controllable activities. *Chem. Commun.* **2014**, *50*, 15305–15308.
- (38) Kerckhoffs, A.; Langton, M. J. Reversible photo-control over transmembrane anion transport using visible-light responsive supramolecular carriers. *Chem. Sci.* **2020**, *11*, 6325–6331.
- (39) Ahmad, M.; Metya, S.; Das, A.; Talukdar, P. A Sandwich Azobenzene–Diamide Dimer for Photoregulated Chloride Transport. *Chem. - Eur. J.* **2020**, *26*, 8703–8708.
- (40) Khairutdinov, R. F.; Hurst, J. K. Light-Driven Transmembrane Ion Transport by Spiropyran–Crown Ether Supramolecular Assemblies. *Langmuir* **2004**, *20*, 1781–1785.
- (41) Salunke, S. B.; Malla, J. A.; Talukdar, P. Phototriggered Release of a Transmembrane Chloride Carrier from an *o*-Nitrobenzyl-Linked Procarrier. *Angew. Chem., Int. Ed.* **2019**, *58*, S354–S358.
- (42) Bao, C.; Ma, M.; Meng, F.; Lin, Q.; Zhu, L. Efficient synthetic supramolecular channels and their light-deactivated ion transport in bilayer lipid membranes. *New J. Chem.* **2015**, *39*, 6297–6302.
- (43) Spörlein, S.; Carstens, H.; Satzger, H.; Renner, C.; Behrendt, R.; Moroder, L.; Tavan, P.; Zinth, W.; Wachtveitl, J. Ultrafast spectroscopy reveals subnanosecond peptide conformational dynamics and validates molecular dynamics simulation. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 7998–8002.
- (44) van Dijken, D. J.; Kovářček, P.; Ihrig, S. P.; Hecht, S. Acylhydrazones as Widely Tunable Photoswitches. *J. Am. Chem. Soc.* **2015**, *137*, 14982–14991.
- (45) Landge, S. M.; Aprahamian, I. A pH Activated Configurational Rotary Switch: Controlling the *E/Z* Isomerization in Hydrazones. *J. Am. Chem. Soc.* **2009**, *131*, 18269–18271.
- (46) Cvrtila, L.; Fanlo-Virgós, H.; Schaeffer, G.; Monreal Santiago, G.; Otto, S. Redox Control over Acyl Hydrazone Photoswitches. *J. Am. Chem. Soc.* **2017**, *139*, 12459–12465.
- (47) Aprahamian, I. Hydrazone switches and things in between. *Chem. Commun.* **2017**, *53*, 6674–6684.
- (48) Su, X.; Aprahamian, I. Hydrazone-based switches, metallo-assemblies and sensors. *Chem. Soc. Rev.* **2014**, *43*, 1963–1981.
- (49) Chaur, M. N.; Collado, D.; Lehn, J. M. Configurational and constitutional information storage: multiple dynamics in systems based on pyridyl and acyl hydrazones. *Chem. - Eur. J.* **2011**, *17*, 248–258.
- (50) Kokan, Z.; Chmielewski, M. J. A Photoswitchable Heteroditopic Ion-Pair Receptor. *J. Am. Chem. Soc.* **2018**, *140*, 16010–16014.
- (51) Vantomme, G.; Lehn, J.-M. Photo- and Thermoresponsive Supramolecular Assemblies: Reversible Photorelease of K^+ Ions and Constitutional Dynamics. *Angew. Chem., Int. Ed.* **2013**, *52*, 3940–3943.
- (52) Leigh, D. A.; Marcos, V.; Nalbantoglu, T.; Vitorica-Yrezabal, I. J.; Yasar, F. T.; Zhu, X. Pyridyl-Acyl Hydrazone Rotaxanes and Molecular Shuttles. *J. Am. Chem. Soc.* **2017**, *139*, 7104–7109.
- (53) Zhou, Y.; Chen, Y.; Zhu, P.-P.; Si, W.; Hou, J.-L.; Liu, Y. Reversible Photo-Gated Transmembrane Channel Assembled from Acylhydrazone-Contained Crown Ether Triad. *Chem. Commun.* **2017**, *53*, 3681–3684.
- (54) Lipinski, C. A.; Lombardo, F.; Dominy, B. W.; Feeney, P. J. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv. Drug Delivery Rev.* **2001**, *46*, 3–26.
- (55) Marvin, ver. 5.8.0; ChemAxon, 2012. <http://www.chemaxon.com>.

- (56) Rasheed, O. K.; Bailey, P. D.; Lawrence, A.; Quayle, P.; Raftery, J. A Modular Synthesis of Multidentate S-, N- and O-Containing Meta- and Paracyclophanes. *Chem. - Eur. J.* **2015**, *2015*, 6988–6993.
- (57) Nakatake, D.; Yokote, Y.; Matsushima, Y.; Yazaki, R.; Ohshima, T. A highly stable but highly reactive zinc catalyst for transesterification supported by a bis(imidazole) ligand. *Green Chem.* **2016**, *18*, 1524–1530.
- (58) *BindFit*, <http://app.supramolecular.org/bindfit/> (accessed July 2017).
- (59) Thordarson, P. Determining association constants from titration experiments in supramolecular chemistry. *Chem. Soc. Rev.* **2011**, *40*, 1305–1323.
- (60) Roy, A.; Saha, T.; Gening, M. L.; Titov, D. V.; Gerbst, A. G.; Tsvetkov, Y. E.; Nifantiev, N. E.; Talukdar, P. Trimodal Control of Ion-Transport Activity on Cyclo-oligo-(1→6)- β -D-glucosamine-Based Artificial Ion-Transport Systems. *Chem. - Eur. J.* **2015**, *21*, 17445–17452.
- (61) Schottel, B. L.; Chifotides, H. T.; Dunbar, K. R. Anion- π interactions. *Chem. Soc. Rev.* **2008**, *37*, 68–83.
- (62) Wu, X.; Howe, E. N. W.; Gale, P. A. Supramolecular Transmembrane Anion Transport: New Assays and Insights. *Acc. Chem. Res.* **2018**, *51*, 1870–1879.
- (63) Davis, J. T.; Gale, P. A.; Quesada, R. Advances in anion transport and supramolecular medicinal chemistry. *Chem. Soc. Rev.* **2020**, *49*, 6056–6086.